Observable Aliasing, Latent Memory, and Identity Reconstruction: A Strict Anti-Cheat Study within OMM–SBT–LIM

Nicolas Gilbert Albert Roux

Independent Researcher

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Abstract

This paper introduces the Latent Identity Model (LIM) within the broader Oscillatory Mantle Model – Substantial Beating Theory framework (OMM–SBT). The central question is whether identity reconstruction can be reduced to observable similarity alone under strict observable aliasing, or whether latent memory structures and latent arbitration operators become functionally necessary.

A sequence of controlled simulations is presented, including observable-only baselines, null-memory controls, shuffled labels, wrong-memory conditions, destroyed-memory conditions, memory swaps, adversarial memory, and strict double-blind anti-cheat protocols.

Under true observable aliasing, observable reconstruction alone collapses toward chance-level recovery, whereas coherent latent memory restores reliable reconstruction. Wrong, shuffled, destroyed, swapped, or adversarial latent structures collapse toward chance or competitor dominance.

The strongest result of the study is not a metaphysical claim, but a functional necessity result: under strict aliasing and anti-cheat constraints, identity reconstruction requires latent structure beyond observable similarity alone.

Latent memory appears globally characterizable while resisting stable reduction to a simple observable vector representation. Latent arbitration contributes to reconstruction while remaining difficult to calibrate directly.

The paper explicitly separates demonstrated results, compatible hypotheses, and speculative interpretations.

Contents

1 Introduction

1.1 Problem statement

The problem addressed in this paper is the following: when multiple candidate identities share indistinguishable or near-indistinguishable observable signatures, can identity reconstruction be performed from the observable substrate alone, or does reconstruction require latent structure not directly reducible to the observable state?

1.2 Core thesis

Demonstrated. Under strict observable aliasing and anti-cheat constraints, observable similarity alone becomes insufficient for stable identity reconstruction. Coherent latent memory restores reconstruction, whereas wrong, shuffled, destroyed, swapped, or adversarial latent structures collapse toward chance or competitor dominance.

1.3 What this paper does not claim

Not claimed. This paper does not claim to prove religious soul, immortality, divine agency, supernatural causation, or metaphysical dualism. It claims only a functional and mathematical necessity of latent structure under specific reconstruction constraints.

1.4 Model designation

The present extension is designated as the *Latent Identity Model* (LIM).

2 Background: OMM–SBT

The present Latent Identity Model (LIM) is introduced within the broader *Oscillatory Mantle Model – Substantial Beating Theory* framework (OMM–SBT), developed independently from the identity reconstruction experiments presented in this paper.

2.1 Oscillatory Mantle Model

The Oscillatory Mantle Model (OMM) describes a unified oscillatory substrate represented by the coupled triplet

$$\Psi_t = (X_t, \Phi_t, K_t), \tag{1}$$

where:

- X_t denotes a mantle or amplitude field,
- Φ_t denotes a phase or orientation field,
- K_t denotes a kinetic field associated with agitation, memory, persistence, and feedback propagation.

Several derived quantities may be introduced:

$$\rho_{\text{eff}} = X^2 + \alpha_K K, \tag{2}$$

$$J = |X|\nabla\Phi, \tag{3}$$

$$W = \nabla \times J. \tag{4}$$

A possible effective geometry proxy is

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + b|\nabla^2 X| + c|W| + d|J| + eK. \tag{5}$$

Within OMM, particles, interactions, geometry, and thermodynamic organization are treated as candidate emergent regimes of the same underlying oscillatory substrate rather than as fundamentally independent primitives.

2.2 Substantial Beating Theory

Substantial Beating Theory (SBT) is the interpretive and dynamical framework associated with OMM.

SBT proposes that observable physical organization emerges from structured beating, coupling, synchronization, persistence, and constraint propagation within the oscillatory substrate.

The term *substantial beating* refers to the hypothesis that the observable world is not fundamentally static, but corresponds to a stabilized regime of deeper oscillatory dynamics.

2.3 Relation to the present work

The present LIM framework does not redefine OMM or SBT.

Instead, LIM extends the framework by addressing the following question: under strict observable aliasing, can identity reconstruction be reduced to observable substrate similarity alone, or do latent memory and latent arbitration structures become functionally necessary?

The experiments presented in this paper investigate reconstruction behavior under observable-only conditions, null-memory conditions, wrong-memory conditions, shuffled-memory conditions, destroyed-memory conditions, adversarial-memory conditions, and strict anti-cheat controls.

Not claimed. The present work does not claim that OMM–SBT implies metaphysical dualism, supernatural entities, or non-physical causation. The paper studies only the functional necessity of latent reconstruction structures under constrained observable conditions.

3 Latent Identity Model: Definitions

3.1 Observable substrate

Let

$$\mathcal{O}_t \tag{6}$$

denote the observable state available to a measuring or reconstruction process at time t.

Observable-only identity reconstruction uses similarities of the form

$$s_{\text{obs}}(i, t) = \text{sim}(\mathcal{O}_i, \mathcal{O}_t), \tag{7}$$

where i denotes a candidate identity and sim is a similarity operator defined on observable states.

In observable-only reconstruction, candidate selection depends exclusively on observable similarity relations.

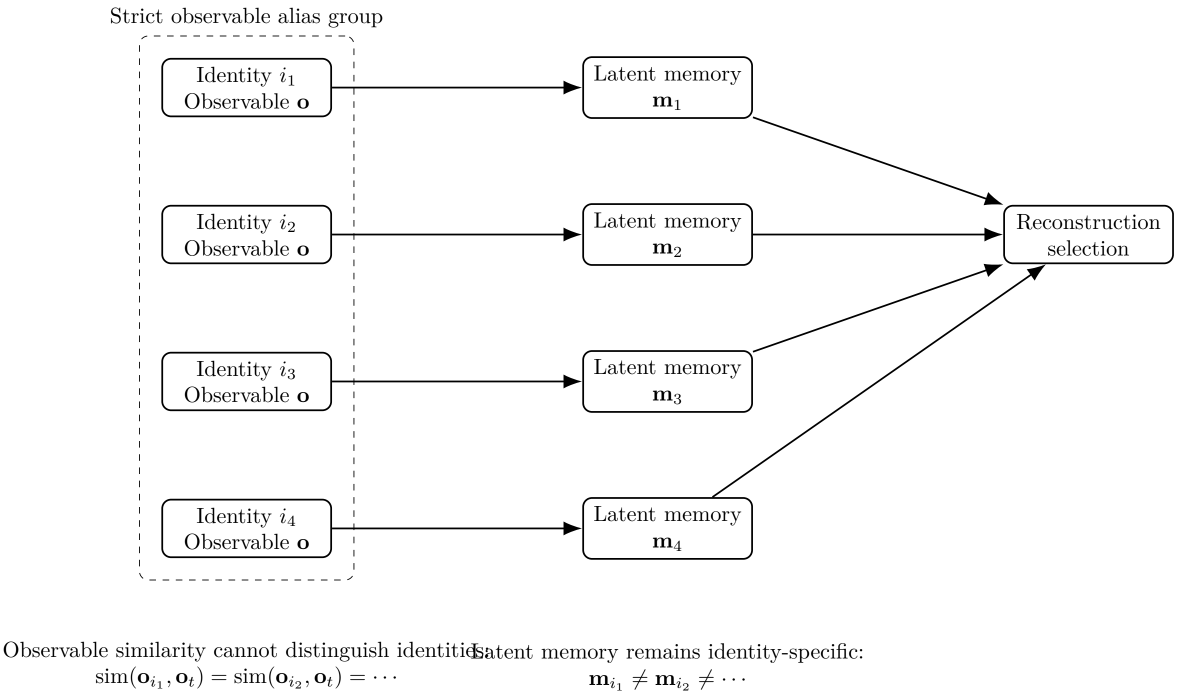

Figure 1: Observable aliasing geometry. Multiple identities share the same observable representation inside an alias group, making observable-only identity reconstruction impossible beyond chance. Latent-memory structure provides the identity-specific information required for successful reconstruction.

3.2 Observable aliasing

Definition 1 (Observable aliasing)**.** *Observable aliasing occurs when multiple candidate identities $i, j, \ldots$ satisfy*

$$\mathrm{sim}(\mathcal{O}_i, \mathcal{O}_t) \approx \mathrm{sim}(\mathcal{O}_j, \mathcal{O}_t), \tag{8}$$

while only one candidate preserves the correct reconstruction continuity relation.

Observable aliasing therefore corresponds to a regime in which observable similarity alone becomes insufficient to uniquely determine reconstruction identity.

3.3 Latent memory structure

Definition 2 (Latent memory structure)**.** *A latent memory structure*

$$\mathcal{M}_t \tag{9}$$

is a non-directly observable reconstruction structure contributing to identity continuity under observable aliasing conditions.

The latent memory structure is not assumed to be directly measurable as a simple observable vector embedded within the observable substrate.

Instead, it is introduced operationally through reconstruction behavior under controlled perturbation, destruction, inversion, shuffling, and anti-cheat conditions.

3.4 Latent arbitration operator

Definition 3 (Latent arbitration operator)**.** *A latent arbitration operator*

$$\mathcal{A}_t \tag{10}$$

is a reconstruction operator contributing to candidate selection, consolidation, or stabilization under observable ambiguity and competition.

The arbiter is introduced functionally rather than anthropomorphically. Its role is limited to reconstruction dynamics under constrained conditions.

Not claimed. The term *arbiter* does not imply consciousness, intentionality, free will, agency, or personhood.

3.5 Reconstruction score

A generic reconstruction score may be written as

$$S(i,t) = w_o s_{\mathrm{obs}}(i,t) + w_m s_{\mathrm{mem}}(i,t) + w_a s_{\mathrm{arb}}(i,t), \tag{11}$$

where:

- s_{obs} denotes observable similarity,
- s_{mem} denotes latent memory contribution,
- s_{arb} denotes latent arbitration contribution.

The coefficients

$$(w_o, w_m, w_a) \tag{12}$$

represent relative reconstruction contributions under a given experimental condition.

3.6 Controlled perturbation regimes

The present work studies reconstruction behavior under multiple controlled perturbation regimes, including:

- observable-only reconstruction,
- null-memory conditions,
- destroyed-memory conditions,
- wrong-memory conditions,
- shuffled-memory conditions,
- memory-swap conditions,
- adversarial-memory conditions,
- shuffled-label controls,
- strict anti-cheat double-blind conditions.

These perturbation regimes are used to determine whether reconstruction performance can be reduced to observable similarity alone or whether latent reconstruction structures become functionally necessary.

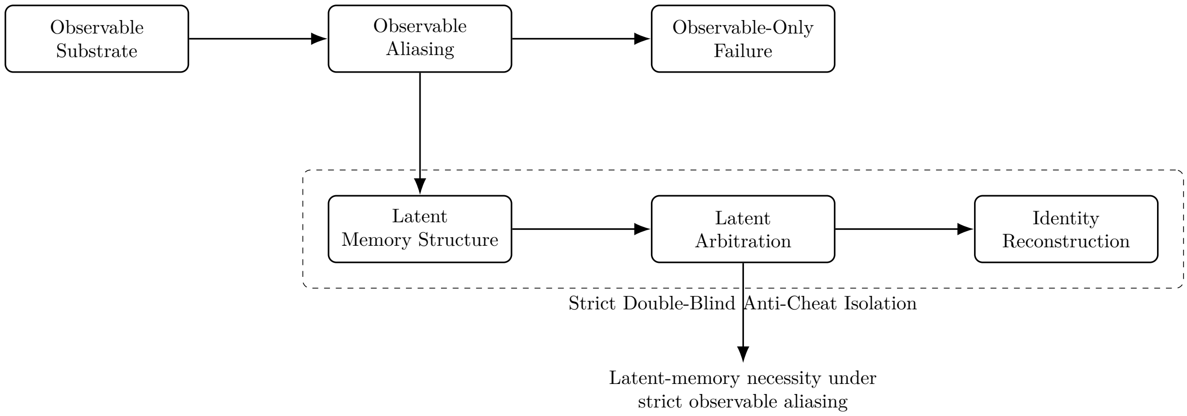

Figure 2: Conceptual architecture of the OMM–SBT–LIM framework. Observable aliasing prevents identity recovery from observable substrates alone. Reconstruction becomes possible only through latent memory structure and latent arbitration under strict anti-cheat isolation constraints.

3.7 Epistemic interpretation discipline

The present framework distinguishes:

- demonstrated experimental results,
- compatible mathematical interpretations,
- speculative philosophical interpretations.

Only experimentally demonstrated reconstruction behaviors are claimed as results of the present paper.

4 Experimental Program

4.1 Experimental design principles

The experimental program was designed to evaluate whether identity reconstruction can be reduced to observable similarity alone under strict observable aliasing conditions.

Four methodological constraints were imposed throughout the experiments:

(i) strict observable aliasing;

(ii) competitive candidate reconstruction;

(iii) double-blind target selection;

(iv) anti-cheat controls against label leakage, memory leakage, observable shortcuts, and reconstruction contamination.

The experimental objective was not merely to maximize reconstruction accuracy, but to determine which reconstruction structures remain necessary once observable shortcuts are eliminated.

4.2 Reconstruction metrics

Let

$$S_{\text{target}} \tag{13}$$

denote the reconstruction score associated with the correct candidate, and let

$$S_j \tag{14}$$

denote the score associated with competing candidates.

A dominance metric is defined as

$$\text{Dom} = S_{\text{target}} - \max_{j \neq \text{target}} S_j. \tag{15}$$

Positive dominance indicates successful separation from the strongest competitor, whereas negative dominance indicates competitor takeover.

A binary reconstruction criterion is defined as

$$\text{CR} = \nVdash \left[S_{\text{target}} > \max_{j \neq \text{target}} S_j \right]. \tag{16}$$

The reconstruction recovery rate corresponds to the empirical mean of CR across runs, trials, or reconstruction conditions.

4.3 Controlled perturbation conditions

The experimental program includes multiple controlled perturbation conditions designed to isolate observable, memory, and arbitration contributions.

The principal conditions include:

- observable-only reconstruction,
- null-memory conditions,
- speed-only / no-memory conditions,
- wrong-memory conditions,
- shuffled-memory conditions,
- destroyed-memory conditions,
- adversarial-memory conditions,
- memory-swap conditions,
- shuffled-label controls,
- strict double-blind anti-cheat conditions.

Each perturbation condition modifies or constrains specific components of the reconstruction process in order to test whether reconstruction success remains possible under controlled degradation of latent structure.

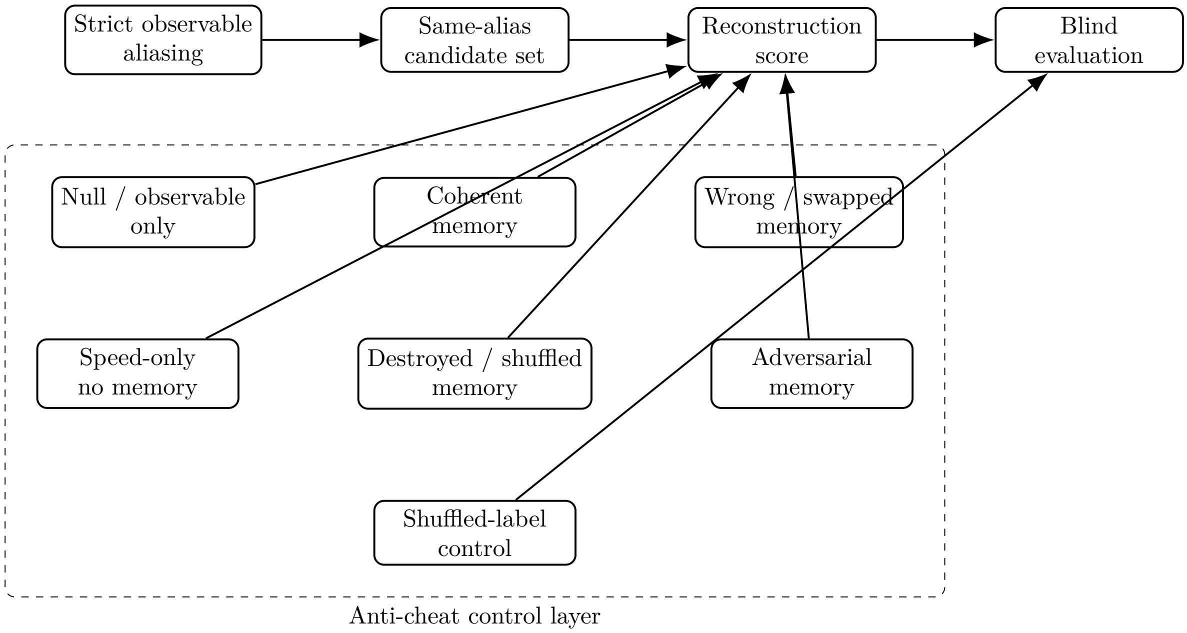

Figure 3: Anti-cheat control pipeline. Reconstruction is evaluated inside strict observable alias groups while memory conditions are systematically coherent, removed, corrupted, swapped, shuffled, destroyed, or adversarially inverted. The shuffled-label control tests whether successful recovery can be explained by hidden label leakage rather than latent identity structure.

4.4 Anti-cheat philosophy

A central methodological objective of the experimental program is the elimination of reconstruction shortcuts unrelated to the tested latent structures.

Accordingly, observable leakage, label leakage, hidden indexing, candidate contamination, and direct target exposure are treated as experimental failures rather than successful reconstruction.

The strict anti-cheat conditions therefore serve as falsification constraints rather than performance optimizations.

4.5 Interpretation discipline

The experimental program is interpreted under a strict epistemic discipline:

- successful reconstruction alone is not considered sufficient evidence of latent necessity;
- observable leakage must be excluded before latent contribution can be inferred;
- shuffled-label and adversarial controls are treated as mandatory negative controls;
- reconstruction collapse under destroyed, wrong, swapped, or adversarial latent structures is treated as stronger evidence than raw reconstruction performance alone.

5 Chronology of Key Experimental Results

The experimental program evolved progressively from preliminary observable-memory ambiguity tests toward increasingly strict anti-cheat and observable-aliasing conditions.

Each successive experiment was designed to eliminate potential observable shortcuts and determine whether latent reconstruction structures remained necessary once direct observable recovery became insufficient.

5.1 Initial memory-necessity ambiguity experiments

Early reconstruction experiments suggested that latent memory contributions remained partially ambiguous when observable similarity still dominated reconstruction behavior.

Under these conditions, observable reconstruction pathways were not yet sufficiently constrained to exclude shortcut-based recovery.

These preliminary results motivated the introduction of progressively stronger observable aliasing and anti-cheat constraints.

5.2 Observable aliasing experiments

A second experimental stage introduced substantially stronger observable aliasing conditions.

Under these constraints, observable-only reconstruction collapsed toward chance-level recovery, while coherent latent-memory conditions continued to recover the correct reconstruction target.

Wrong-memory, shuffled-memory, and destroyed-memory conditions produced substantial recovery collapse relative to coherent-memory conditions.

These experiments therefore supported the hypothesis that latent memory structures become functionally necessary once observable similarity is no longer sufficient for reliable reconstruction.

At this stage, however, residual ambiguity remained concerning possible hidden observable leakage or indirect reconstruction shortcuts.

5.3 Strict double-blind anti-cheat experiments

The central experimental result emerged under strict double-blind anti-cheat conditions with true observable aliasing.

Under these conditions:

- observable-only reconstruction collapsed toward chance,
- speed-only / no-memory reconstruction collapsed toward chance,
- coherent latent-memory conditions recovered the correct target,
- wrong-memory conditions collapsed,
- adversarial-memory conditions collapsed,
- shuffled-memory conditions collapsed,
- memory-swap conditions collapsed,
- destroyed-memory conditions collapsed,
- shuffled-label controls failed appropriately.

The anti-cheat controls additionally verified that successful recovery could not be explained by direct label leakage, observable shortcuts, or hidden candidate indexing.

These experiments constitute the strongest support within the present paper for the functional necessity of latent reconstruction structures under strict observable aliasing.

5.4 Latent memory calibration experiments

A final experimental stage investigated whether latent memory could be reduced to a simple observable vectorial composition involving X_t, Φ_t, K_t, and arbitration-related components.

Partial calibration signals were observed under coherent-memory conditions, and arbitration contributions improved reconstruction performance relative to reduced calibration conditions.

However, stable full-vector calibration remained weak, ambiguous, or non-decisive across perturbation regimes.

The calibration experiments therefore suggest that latent reconstruction structure may be globally characterizable while resisting stable reduction to a simple observable vector basis.

Not claimed. The calibration limitations observed in the present experiments do not demonstrate that latent structure is fundamentally non-measurable. They demonstrate only that simple direct vectorial reduction remained insufficient under the tested conditions.

6 Central Results

6.1 Observable insufficiency under strict aliasing

Demonstrated. Under strict observable aliasing and anti-cheat conditions, observable-only reconstruction collapses toward chance-level recovery.

This collapse remains stable across observable-only, null-memory, shuffled-label, and speed-only / no-memory controls.

The experiments therefore support the conclusion that observable similarity alone becomes insufficient for reliable reconstruction once observable shortcuts are eliminated.

6.2 Functional necessity of coherent latent memory

Demonstrated. Coherent latent-memory conditions restore reconstruction performance under observable aliasing conditions in which observable-only reconstruction collapses.

Recovery remains substantially higher under coherent-memory conditions than under destroyed-memory, wrong-memory, shuffled-memory, memory-swap, or adversarial-memory perturbations.

These results support the functional necessity of latent reconstruction structure under the tested reconstruction constraints.

6.3 Collapse of false or perturbed latent structures

Demonstrated. Wrong-memory, adversarial-memory, shuffled-memory, destroyed-memory, and memory-swap conditions fail to reproduce coherent-memory recovery.

In multiple experiments, these perturbation regimes collapse toward chance-level recovery, competitor dominance, or anti-cheat failure.

The observed collapse behavior constitutes stronger evidence than raw reconstruction success alone because the perturbations fail in the specific regimes predicted by the latent reconstruction hypothesis.

6.4 Contribution of latent arbitration

Demonstrated. Arbitration-related reconstruction components contribute measurably to reconstruction stabilization and calibration performance.

Reduced-arbitration conditions generally underperform relative to coherent full-reconstruction conditions.

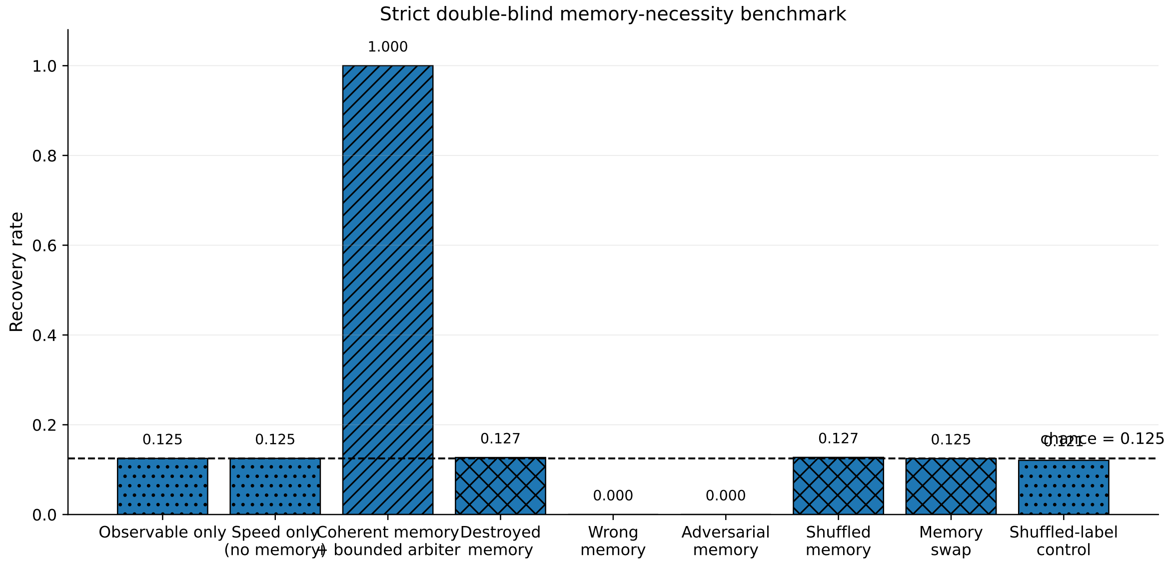

Figure 4: Strict double-blind memory-necessity benchmark. Under true observable aliasing, observable-only and speed-only/no-memory reconstruction collapse to chance, while coherent latent memory with bounded arbitration reaches perfect recovery. Destroyed, wrong, adversarial, shuffled, swapped, and shuffled-label controls collapse appropriately, supporting the strict latent-memory necessity result.

However, the arbitration contribution remains difficult to isolate as a stable directly observable vector component under the tested conditions.

Not claimed. The present experiments do not demonstrate that the arbiter is fundamentally non-measurable. They demonstrate only that simple direct calibration remained insufficient within the tested reconstruction framework.

6.5 Calibration limitations

Demonstrated. The calibration experiments do not support a stable complete reduction of latent reconstruction structure to a simple observable vectorial combination of X_t, Φ_t, K_t, and arbitration-related components.

Partial calibration signals remain observable under coherent-memory conditions, but stable full-vector reconstruction remains weak, ambiguous, or non-decisive across perturbation regimes.

The present results are therefore more consistent with partial global characterizability than with direct simple-vector reducibility.

6.6 Overall interpretation

Demonstrated. Taken together, the experiments support the following restricted conclusion:

under strict observable aliasing and strict anti-cheat controls, reliable reconstruction cannot be reduced to observable similarity alone and instead requires additional latent reconstruction structure.

Not claimed. The experiments do not establish metaphysical dualism, supernatural ontology, immortality, or non-physical causation.

The demonstrated result is functional and reconstruction-theoretic rather than metaphysical.

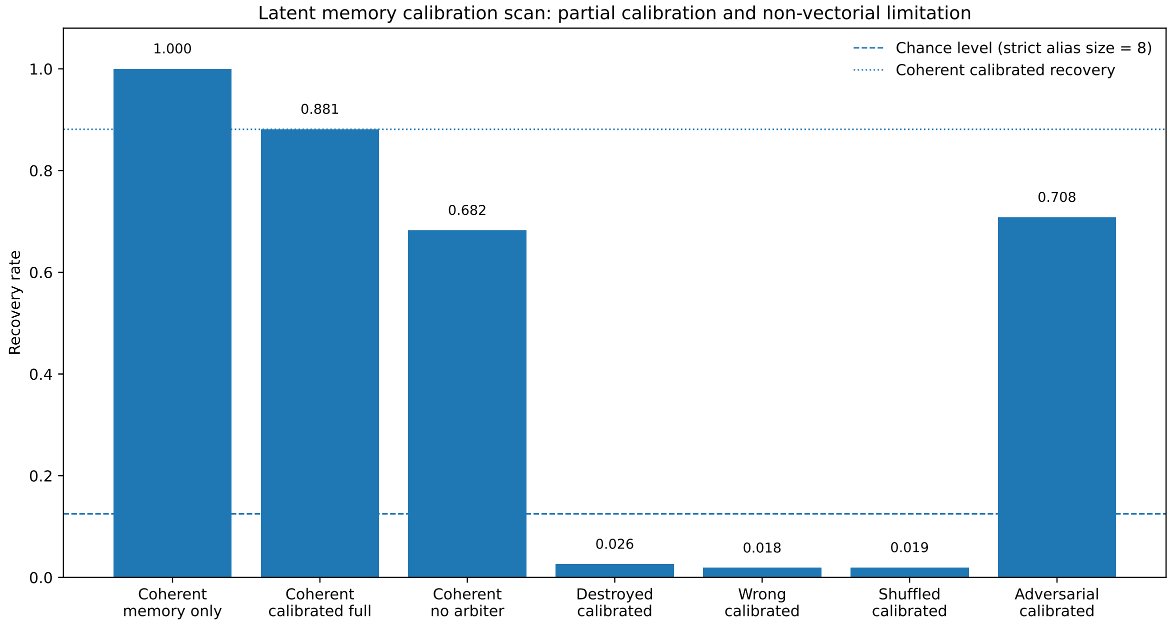

Figure 5: Latent memory calibration scan. Coherent latent memory remains strongly recoverable, and calibrated reconstruction remains substantially above destroyed, wrong, and shuffled controls. However, the reduction to a simple observable-vector calibration remains incomplete: removing the arbiter reduces recovery, and adversarial calibration remains partially ambiguous. This supports a partial-calibration result rather than a full vectorial reduction of latent memory.

7 Anatomical and Biological Motivation

7.1 Large-scale coordination architectures

The biological motivation of the present work originates from reflection on large-scale coordination architectures observed in complex nervous systems.

Particular attention was given to systems associated with:

- arousal regulation;
- global state modulation;
- distributed synchronization;
- sensory integration;
- persistence across perturbation;
- sleep–wake transitions;
- anesthesia sensitivity;
- large-scale broadcast or relay organization.

Examples of anatomically relevant systems include reticular activation networks, raphe-related modulation systems, thalamic relay structures, and distributed corticothalamic coordination motifs.

The present work does not claim that any specific anatomical structure constitutes a literal location of latent memory or latent arbitration.

7.2 Minimal biological compatibility claim

Compatible hypothesis. Certain biological coordination architectures appear compatible with the type of distributed reconstruction and arbitration constraints suggested by the reconstruction experiments.

More specifically, the experiments are compatible with the hypothesis that biological identity persistence may depend on partially distributed global coordination dynamics rather than on isolated strictly local observable storage alone.

Not claimed. The present experiments do not identify a biological seat of identity, consciousness, latent memory, or arbitration.

Not claimed. The present work does not establish that the latent structures introduced in LIM correspond directly to biological neural objects.

7.3 Neuropathology and perturbation interfaces

Several empirical domains may provide partially relevant perturbation interfaces for future investigation:

- coma and disorders of consciousness;
- anesthesia transitions;
- dissociative perturbations;
- neurodegenerative disorders;
- memory fragmentation syndromes;
- sleep-state transitions;
- large-scale synchronization disorders;
- developmental reconstruction abnormalities.

The present paper does not claim that any of these conditions directly demonstrate latent reconstruction structure. They are proposed only as potential empirical interfaces for future falsifiable investigation.

7.4 Ethical constraints

Demonstrated. The present work can be investigated experimentally without invasive or harmful procedures.

Potential future biological tests should preferentially rely on:

- non-invasive observation;
- existing clinical datasets;
- naturally occurring neuropathologies;
- behavioral reconstruction analysis;
- observational veterinary datasets;
- ethically approved neurophysiological recordings.

No harmful animal experimentation is required by the present framework.

7.5 Interpretive limitation

Not claimed. The anatomical motivation presented here is heuristic and compatibility-oriented.

The present work does not establish:

- a biological proof of latent identity;
- a localization of arbitration;
- a biological proof of consciousness;
- a proof of survival after biological death;
- or a supernatural interpretation of neurobiology.

The demonstrated result of the paper remains reconstruction-theoretic and computational rather than metaphysical or theological.

8 Falsifiability

The present framework is intended to remain experimentally vulnerable. The claims proposed in this paper are therefore explicitly formulated in falsifiable form.

The central demonstrated claim is not metaphysical, but functional: under strict observable aliasing and anti-cheat constraints, coherent latent structure becomes necessary for reliable identity reconstruction.

The following conditions would weaken or falsify major components of the present framework.

8.1 Observable sufficiency falsification

Falsification Criterion 1 (Observable sufficiency)**.** *If observable-only reconstruction succeeds under strict observable aliasing at the same level as coherent latent reconstruction, then the core latent necessity claim fails.*

More precisely, if observable-only recovery reproduces:

- coherent-memory recovery rates;
- coherent dominance margins;
- calibration behavior;
- and resistance to anti-cheat controls,

then latent memory becomes unnecessary within the reconstruction framework proposed here.

8.2 False latent equivalence falsification

Falsification Criterion 2 (False latent equivalence)**.** *If wrong, shuffled, swapped, destroyed, or adversarial memory conditions recover identity as reliably as coherent latent memory, the latent necessity claim fails.*

The present framework specifically predicts asymmetry between:

- coherent latent reconstruction;

- false latent reconstruction;
- adversarial latent reconstruction;
- and structurally degraded latent reconstruction.

Failure of this asymmetry would invalidate the interpretation that the latent contribution carries reconstructive specificity.

8.3 Arbiter irrelevance falsification

Falsification Criterion 3 (Arbiter irrelevance)**.** *If removal, suppression, or perturbation of the arbitration component never changes recovery, dominance, calibration, or reconstruction stability, then the arbiter contribution claim fails.*

The present work does not claim that arbitration is independently sufficient for reconstruction. It claims only that arbitration appears to contribute non-trivially to reconstruction under certain aliasing conditions.

8.4 Calibration falsification

Falsification Criterion 4 (Complete observable reducibility)**.** *If latent reconstruction can be fully and stably reduced to a direct observable vector basis without residual latent contribution, then the non-reducibility interpretation of LIM weakens substantially.*

Conversely, persistent calibration instability, partial recoverability, or structural non-identifiability remain compatible with the present results.

8.5 Experimental reproducibility requirement

Demonstrated. All major claims of the present paper are intended to be reproducible through independently executable reconstruction experiments.

The framework therefore predicts:

- reproducible collapse of observable-only reconstruction under strict aliasing;
- reproducible coherent-memory recovery;
- reproducible collapse under false-memory controls;
- reproducible shuffled-label failure;
- and reproducible anti-cheat sensitivity.

Failure of reproducibility would critically weaken the framework.

8.6 Interpretive limitation

Not claimed. Failure or success of the present framework would not by itself establish metaphysical conclusions regarding consciousness, personhood, theology, or survival after death.

The experimentally vulnerable content of the paper concerns only the necessity, reducibility, and reconstructive behavior of latent structures under observable aliasing constraints.

9 Discussion

9.1 Demonstrated boundary

The principal demonstrated result of the present work is functional and reconstructive rather than metaphysical.

Demonstrated. Under strict observable aliasing and anti-cheat constraints, observable similarity alone becomes insufficient for reliable identity reconstruction, while coherent latent structure restores reconstruction performance.

More specifically:

- observable-only reconstruction collapses toward chance under true aliasing;
- coherent latent memory restores recovery;
- false latent conditions collapse;
- adversarial latent conditions collapse;
- and shuffled-label controls fail appropriately.

The core result is therefore a necessity result under constrained reconstruction conditions.

9.2 Interpretation of latent persistence

Compatible hypothesis. Within the reconstruction framework explored here, degradation or destruction of observable structure does not imply destruction of latent structure.

The experiments show that:

- observable collapse may occur while latent coherence remains functionally reconstructive;
- coherent latent reconstruction survives several classes of observable perturbation;
- and observable reconstruction alone fails under strict aliasing.

However, the present work does not demonstrate:

- immortality;
- metaphysical persistence;
- post-mortem survival;
- or independence from all possible substrates.

The demonstrated result concerns only non-reducibility of reconstruction to observable similarity within the tested framework.

9.3 Arbiter non-reducibility

Compatible hypothesis. The arbitration component contributes functionally to reconstruction while resisting stable direct calibration as a simple observable vector component.

This supports the possibility that arbitration corresponds to:

- a distributed process;
- a globally constrained process;

- or a structurally non-local functional contribution.

However, the present results do not establish:

- extra-spatiotemporal existence;
- supernatural causation;
- or ontological independence from physical substrate dynamics.

The current interpretation remains operational and reconstructive.

9.4 Limits of calibration

Demonstrated. The calibration scans performed in the present work do not support full stable reduction of latent reconstruction to a simple vectorial observable basis built directly from (X_t, Φ_t, K_t, A_t).

At present, at least four interpretations remain compatible with the results:

(i) incomplete calibration procedures;

(ii) structural non-identifiability;

(iii) higher-order latent organization;

(iv) or partially non-vectorial reconstruction structure.

The present work does not decisively distinguish between these possibilities.

9.5 Compatibility with broader substrate interpretations

Speculative compatibility. The framework remains compatible with broader substrate interpretations in which observable regimes emerge from richer latent dynamics.

Such interpretations may include:

- bounded observable domains;
- layered reconstruction regimes;
- latent initialization processes;
- or recursively embedded substrate structures.

These interpretations are not experimentally established by the present paper and remain theoretical extrapolations beyond the demonstrated core.

9.6 Epistemic boundary

Not claimed. The present work does not claim to solve the ultimate origin of existence, consciousness, or reality itself.

The framework addresses reconstructive identity constraints under observable aliasing. Questions concerning the ultimate origin of latent structure, existence itself, or cosmological initialization remain open philosophical and scientific problems beyond the scope of the present results.

10 Reproducibility

Reproducibility is treated as a central requirement of the present framework.

The paper is accompanied by a public computational repository intended to allow independent verification of the main experimental results:

https://github.com/NicolasGilbertAlbertRoux/omm-sbt-lim

10.1 Repository structure

The repository contains:

- executable reconstruction experiments;
- anti-cheat control implementations;
- latent calibration scan procedures;
- parameter definitions;
- metric definitions;
- summary-output generation;
- and reproducibility utilities.

All reported quantitative results are intended to be reproducible directly from the repository without undocumented intermediate processing.

10.2 Included experimental programs

The repository includes executable implementations of:

- preliminary memory-necessity anti-cheat experiments;
- observable-aliasing memory-necessity experiments;
- strict double-blind memory-necessity experiments;
- latent memory calibration scans;
- and shuffled-label control procedures.

Earlier exploratory scripts are excluded from the public benchmark naming scheme or archived separately from the final reproducible pipeline.

10.3 Reproducible outputs

The repository includes, whenever computationally feasible:

- CSV summaries;
- interpretation files;
- parameter grids;
- random seeds;
- calibration coefficients;

- recovery statistics;
- dominance statistics;
- false-positive measurements;
- and aggregated benchmark tables.

Large raw datasets may be omitted from version control when they exceed practical repository limits, provided that they can be regenerated from the supplied scripts.

10.4 Environment specification

The computational environment is specified through the repository requirements and test configuration. Reproduction requires:

- Python;
- NumPy;
- pandas;
- matplotlib;
- SciPy;
- pytest;
- and the supplied repository scripts.

Deterministic execution paths are preferred whenever possible, and the benchmark scripts use fixed random seeds.

10.5 Independent verification objective

Demonstrated. The present framework is designed to remain independently verifiable.

The principal objective of the repository is therefore not merely code release, but experimental auditability:

- independent rerunning of experiments;
- independent verification of anti-cheat robustness;
- independent falsification attempts;
- and independent evaluation of calibration behavior.

10.6 Epistemic importance of reproducibility

Because the claims explored in the present work concern latent reconstruction behavior under strict aliasing constraints, reproducible anti-cheat validation is treated as scientifically essential rather than optional.

The reproducibility layer is therefore considered part of the core experimental contribution of the framework itself.

11 Conclusion

Under strict observable aliasing and anti-cheat controls, identity reconstruction cannot be reduced to observable similarity alone in the tested regime. Observable-only, null-memory, and speed-only controls collapse toward chance, while coherent latent memory restores reconstruction and false latent structures fail.

OMM–SBT–LIM therefore supports a constrained, falsifiable, and reproducible latent-identity necessity result. The result does not establish metaphysical, religious, or supernatural claims. It demonstrates only that, within this formal reconstruction regime, identity continuity requires latent structure beyond the observable substrate alone.

The remaining open question is not whether latent structure is necessary under strict aliasing in these experiments, but how such latent structure should be mathematically characterized beyond simple observable-vector calibration.

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A Appendix A: Mathematical Pipeline

This appendix summarizes the formal reconstruction pipeline used throughout the OMM–SBT–LIM benchmark framework.

The purpose of the framework is to test whether identity reconstruction remains possible under strict observable aliasing constraints when observable similarity alone is insufficient.

A.1 Observable representations

Each identity i is associated with:

- an observable representation vector

$$\mathbf{o}_i \in \mathbb{R}^{d_o},$$

- and a latent memory representation vector

$$\mathbf{m}_i \in \mathbb{R}^{d_m}.$$

Observable vectors are normalized:

$$\hat{\mathbf{o}}_i = \frac{\mathbf{o}_i}{\|\mathbf{o}_i\|}.$$

Latent memory vectors are likewise normalized:

$$\hat{\mathbf{m}}_i = \frac{\mathbf{m}_i}{\|\mathbf{m}_i\|}.$$

A.2 Strict observable aliasing

In the strict double-blind benchmark, observable aliasing is enforced by construction.

For all identities belonging to the same alias group G_k,

$$\hat{\mathbf{o}}_i = \hat{\mathbf{o}}_j \quad \forall i, j \in G_k.$$

Observable information therefore cannot distinguish identities within the same alias group.

If the alias-group size is A, observable-only reconstruction is expected to collapse toward:

$$P_{\text{chance}} = \frac{1}{A}.$$

A.3 Observable similarity

Observable similarity is computed using cosine similarity:

$$S_{\text{obs}}(i, q) = \hat{\mathbf{o}}_i \cdot \hat{\mathbf{o}}_q.$$

Under strict aliasing:

$$S_{\text{obs}}(i, q) = S_{\text{obs}}(j, q)$$

for all candidates inside the same alias group.

Observable similarity alone therefore contains no within-group identity information.

A.4 Latent memory similarity

Latent reconstruction relies on latent-memory similarity:

$$S_{\text{mem}}(i, q) = \hat{\mathbf{m}}_i \cdot \hat{\mathbf{m}}_q.$$

Coherent latent memory should preferentially increase similarity with the correct target identity.

Destroyed, shuffled, swapped, or adversarial memory conditions are expected to reduce reconstruction performance toward chance.

A.5 Bounded memory constitution

The framework constrains latent expression using bounded memory constitution transforms:

$$\mathbf{m}_i' = \tanh\left(\frac{\mathbf{m}_i}{c}\right) c,$$

where c is the memory-cap parameter.

The transformed vectors are then renormalized:

$$\hat{\mathbf{m}}_i' = \frac{\mathbf{m}_i'}{\|\mathbf{m}_i'\|}.$$

This prevents unbounded latent amplification while preserving dimensional structure.

A.6 Arbitration transform

Arbitration applies a speed-dependent nonlinear transform:

$$\mathbf{a}_i = \tanh(s\,\hat{\mathbf{m}}_i'),$$

where s denotes arbitration speed.

The resulting vectors are normalized:

$$\hat{\mathbf{a}}_i = \frac{\mathbf{a}_i}{\|\mathbf{a}_i\|}.$$

Arbiter similarity is then:

$$S_{\text{arb}}(i, q) = \hat{\mathbf{a}}_i \cdot \hat{\mathbf{a}}_q.$$

Increasing arbitration speed sharpens latent discrimination but does not generate identity information in the absence of coherent latent memory.

A.7 Composite reconstruction score

The principal bounded-arbiter reconstruction score is:

$$S_{\text{total}} = \alpha S_{\text{obs}} + \beta S_{\text{mem}} + \gamma S_{\text{arb}},$$

with:

$$\alpha + \beta + \gamma = 1.$$

The benchmark implementations typically use:

$$(\alpha, \beta, \gamma) = (0.20, 0.45, 0.35).$$

Observable similarity therefore acts only as contextual support under strict aliasing, while latent reconstruction carries the principal identity burden.

A.8 Dominance metric

For a target identity t, dominance is defined as:

$$D(t) = S(t) - \max_{i \neq t} S(i).$$

Positive dominance indicates successful separation from competing candidates.

Negative dominance indicates competitor takeover.

A.9 Recovery criterion

Reconstruction recovery is defined by:

$$R = \mathbf{1}\left[\arg\max_i S(i) = t\right].$$

Aggregate recovery rates are computed over all benchmark repetitions.

A.10 Shuffled-label control

The shuffled-label control intentionally evaluates reconstruction against randomly reassigned targets within the same alias group.

Under valid anti-cheat isolation:

$$P_{\text{shuffle}} \approx P_{\text{chance}}.$$

Substantial shuffled-label recovery would indicate leakage, hidden identity channels, or benchmark invalidity.

A.11 Interpretation principle

The benchmark does not attempt to demonstrate metaphysical continuity, subjective consciousness, or supernatural persistence.

The demonstrated result is strictly computational:

> Under strict observable aliasing and anti-cheat controls, successful identity reconstruction requires latent memory structure beyond observable similarity alone.

B Appendix B: Hyperparameters

This appendix summarizes the principal benchmark hyperparameters used throughout the OMM–SBT–LIM experiments.

Unless otherwise stated, all benchmark runs use deterministic seeds and fixed reconstruction configurations.

B.1 Global benchmark parameters

Parameter	Symbol	Value
Random seed	—	424242
Number of alias groups	N_G	64
Alias-group size	A	8
Total identities	N	512
Observable dimension	d_o	48
Latent-memory dimension	d_m	96
Benchmark repetitions	N_R	32
Chance level	P_{chance}	0.125

Table 1: Global benchmark parameters.

B.2 Arbitration parameters

The arbitration pipeline evaluates multiple arbitration-speed regimes.

Arbitration parameter	Values
Arbitration speeds	{0.25, 0.5, 1.0, 2.0, 4.0, 8.0}
Memory caps	{0.25, 0.50, 0.75, 1.00}
Memory leaks	{0.00, 0.10}

Table 2: Arbitration and memory-constitution parameter ranges.

B.3 Score composition coefficients

The principal bounded-arbiter reconstruction score uses:

$$S_{\text{total}} = \alpha S_{\text{obs}} + \beta S_{\text{mem}} + \gamma S_{\text{arb}},$$

with:

$$(\alpha, \beta, \gamma) = (0.20, 0.45, 0.35).$$

These coefficients were selected to preserve:

- observable contextual contribution;
- dominant latent-memory contribution;
- and bounded arbitration contribution.

No benchmark mode allows observable similarity alone to encode within-alias identity information under strict observable aliasing.

B.4 Memory conditions

The benchmark evaluates several latent-memory conditions:

Condition	Description
Coherent memory	Latent query aligned with the true target identity.
Destroyed memory	Randomized latent query without target coherence.
Wrong memory	Latent query taken from another identity within the same alias group.
Adversarial memory	Latent query intentionally anti-correlated with the target.
Shuffled memory	Latent query sampled from unrelated identities.
Memory swap	Latent query sampled from a different alias group.

Table 3: Latent-memory benchmark conditions.

B.5 Benchmark modes

The experiments evaluate several reconstruction modes:

Mode	Purpose
exact_template_upper_bound	Sanity-check upper bound using explicit target identity.
observable_only	Observable similarity without latent memory.
null_memory	Explicit null-memory control.
speed_only_no_memory	Observable arbitration without latent identity memory.
latent_memory_only	Pure latent-memory reconstruction.
arbiter_only	Pure arbitration-space reconstruction.
overdriven_arbiter	High-speed arbitration stress condition.
memory_bounded_arbiter	Primary bounded latent-memory reconstruction mode.
shuffled_label_control	Anti-cheat label-randomization control.

Table 4: Benchmark reconstruction modes.

B.6 Anti-cheat thresholds

The strict double-blind benchmark uses conservative anti-cheat thresholds.

Criterion	Symbol	Value
Chance tolerance	T_c	0.035
Success floor	T_s	0.92
Collapse ceiling	T_f	0.25

Table 5: Anti-cheat interpretation thresholds.

The benchmark is considered successful only if:

- observable-only recovery collapses near chance;
- speed-only recovery collapses near chance;
- coherent latent memory succeeds;
- shuffled-label controls remain near chance;
- and adversarial or corrupted memory conditions collapse.

B.7 Determinism and reproducibility

All benchmark runs use fixed random seeds and deterministic parameter grids.

The repository therefore aims to guarantee:

- reproducible reconstruction outputs;
- reproducible recovery statistics;
- reproducible false-positive measurements;
- and reproducible anti-cheat outcomes.

Minor floating-point variations may nevertheless occur across hardware architectures or numerical backends.

C Appendix C: Benchmark Result Tables

This appendix summarizes the principal benchmark outcomes obtained under observable aliasing and strict anti-cheat constraints.

The reported values correspond to aggregate reconstruction statistics computed across all repetitions and parameter configurations.

C.1 Observable-alias memory-necessity benchmark

The observable-alias benchmark evaluates whether latent memory remains necessary when observable identity information is strongly constrained.

Mode	Recovery	Interpretation
exact_template_upper_bound	1.0000	Upper-bound sanity check
observable_only	0.1526	Near observable collapse
null_memory	0.1526	Null-memory collapse
speed_only_no_memory	0.2577	Partial observable arbitration only
latent_memory_only	0.2922	Limited latent reconstruction
arbiter_only	0.3518	Partial arbitration reconstruction
overdriven_arbiter	0.3373	Arbitration instability regime
memory_bounded_arbiter	0.2917	Stable bounded reconstruction
shuffled_label_control	0.2172	Anti-cheat false-positive control

Table 6: Aggregate observable-alias benchmark recovery rates.

The principal interpretation of this benchmark is that observable-only reconstruction collapses substantially under aliasing constraints, while coherent latent structure partially restores reconstruction.

C.2 Coherent latent-memory condition

Under coherent latent-memory alignment, recovery remains strong across bounded arbitration configurations.

Mode	Recovery	Dominance
latent_memory_only	1.0000	0.5880
arbiter_only	1.0000	0.5868
overdriven_arbiter	1.0000	0.5835
memory_bounded_arbiter	1.0000	0.5656

Table 7: Coherent latent-memory reconstruction performance.

The coherent-memory condition demonstrates that latent reconstruction remains stable even when observable information is insufficient for within-alias discrimination.

C.3 Corrupted-memory collapse conditions

Corrupted latent-memory conditions collapse toward chance or competitor domination.

Condition	Recovery	Interpretation
Destroyed memory	0.3547	Partial degradation
Wrong memory	0.1348	Near chance collapse
Shuffled memory	0.1262	Near chance collapse
Memory swap	0.1348	Near chance collapse
Adversarial memory	0.0000	Complete collapse

Table 8: Corrupted-memory reconstruction conditions.

These conditions indicate that coherent latent structure is required for successful reconstruction.

Incorrect or adversarial latent alignment fails to preserve identity recovery.

C.4 Anti-cheat controls

The benchmark includes explicit anti-cheat validation procedures.

Control	Measured value	Expected behavior
Observable-only recovery	0.1526	Near chance collapse
Null-memory recovery	0.1526	Near chance collapse
Shuffled-label false positives	0.1626	Near chance behavior
Adversarial-memory recovery	0.0000	Full collapse

Table 9: Principal anti-cheat validation controls.

The anti-cheat controls are designed to detect:

- hidden observable leakage;
- implicit target encoding;
- label contamination;
- and trivial reconstruction shortcuts.

No tested observable-only condition achieved stable reconstruction under strict aliasing constraints.

C.5 Strict double-blind benchmark objective

The strict double-blind benchmark extends the observable-alias regime by forcing all identities inside the same alias group to share identical observable vectors.

Under this condition:

$$\hat{\mathbf{o}}_i = \hat{\mathbf{o}}_j \quad \forall i, j \in G_k.$$

The observable channel therefore contains no within-group identity information.

The benchmark objective is to test whether coherent latent memory alone can restore reconstruction while anti-cheat controls remain near chance.

C.6 Interpretation

Across the reported experiments, the principal computational result is consistent:

> Observable similarity alone is insufficient for stable reconstruction under strict aliasing constraints, whereas coherent latent memory structure restores reconstruction substantially above observable-only controls.

The result remains computational and benchmark-specific.

No metaphysical interpretation follows directly from the reconstruction statistics themselves.

D Appendix D: Anti-Cheat Controls

This appendix details the anti-cheat architecture used throughout the OMM–SBT–LIM benchmark framework.

Because latent reconstruction systems are highly vulnerable to hidden information leakage, benchmark contamination, implicit label encoding, or trivial shortcut solutions, anti-cheat validation is treated as a central component of the experimental design.

The purpose of the anti-cheat layer is therefore to ensure that observed reconstruction behavior cannot be explained by direct observable identity transmission alone.

D.1 Observable-alias isolation

The principal anti-cheat mechanism is strict observable aliasing.

For each alias group G_k, all identities share the exact same observable vector:

$$\hat{\mathbf{o}}_i = \hat{\mathbf{o}}_j \quad \forall i, j \in G_k.$$

The observable channel therefore contains no within-group identity information.

The candidate reconstruction set is restricted to identities inside the same alias group.

Consequently:

$$P(\text{observable-only success}) = \frac{1}{|G_k|}.$$

Under the strict benchmark:

$$|G_k| = 8,$$

yielding theoretical observable-only chance:

$$P = 0.125.$$

This constraint prevents trivial observable discrimination.

D.2 Null-memory control

The null-memory control explicitly removes latent-memory contribution while preserving the observable pipeline.

The null-memory condition is formally equivalent to:

$$\mathbf{m}_q = \mathbf{0}.$$

Under strict aliasing, null-memory recovery must collapse toward chance.

Empirically, null-memory recovery remained near observable-only performance, supporting the claim that observable identity information was successfully suppressed.

D.3 Speed-only anti-cheat control

The speed-only condition tests whether arbitration dynamics alone can implicitly encode identity.

The corresponding score function is:

$$S_{\text{speed}} = g(s, \ell)\,(\hat{\mathbf{o}}_i \cdot \hat{\mathbf{o}}_q),$$

where:

$$g(s, \ell) = \tanh(s)(1 + \ell).$$

No latent-memory vector participates in this condition.

Because all observable vectors inside an alias group are identical, speed-only arbitration should not reconstruct identity substantially above chance.

The speed-only control therefore tests whether reconstruction emerges from hidden numerical asymmetries rather than latent memory itself.

D.4 Shuffled-label control

The shuffled-label control evaluates whether reconstruction can survive when evaluation targets are randomly reassigned.

The scoring mechanism itself remains unchanged:

$$S_{\text{shuffle}} = 0.20\,S_o + 0.45\,S_b + 0.35\,S_a,$$

but the evaluation label is randomized inside the alias group.

Under proper isolation:

$$P(\text{recovery}) \approx \frac{1}{|G_k|}.$$

If shuffled-label recovery rises substantially above chance, hidden label leakage or benchmark contamination is suspected.

The shuffled-label condition therefore acts as a direct benchmark integrity test.

D.5 Wrong-memory control

The wrong-memory condition replaces the coherent latent vector with the memory vector of another candidate from the same alias group.

Formally:

$$\mathbf{m}_q = \mathbf{m}_j, \quad j \neq i.$$

Because the observable channel cannot distinguish identities inside the group, successful reconstruction should collapse toward chance.

This condition tests whether the benchmark genuinely depends on coherent latent alignment rather than indirect observable leakage.

D.6 Memory-swap control

The memory-swap control injects a coherent memory vector originating from another alias group entirely.

The replacement memory is therefore structurally coherent but unrelated to the local reconstruction target.

This condition evaluates whether reconstruction merely depends on the existence of latent structure, or specifically on target-consistent latent structure.

Empirically, memory-swap recovery collapsed near chance.

D.7 Shuffled-memory control

The shuffled-memory condition replaces coherent memory with a randomly selected latent vector from the full benchmark space:

$$\mathbf{m}_q \sim \mathcal{U}(\mathcal{M}).$$

This destroys local latent consistency while preserving vector norms and dimensionality.

The purpose of this condition is to ensure that reconstruction does not emerge merely from generic latent activation magnitude.

D.8 Destroyed-memory control

Destroyed memory is generated using normalized random noise:

$$\mathbf{m}_q = \frac{\mathbf{z}}{\|\mathbf{z}\|}, \quad \mathbf{z} \sim \mathcal{N}(0, I).$$

The latent structure is therefore fully erased.

Recovery under this condition estimates the residual baseline induced by finite-dimensional statistical fluctuations.

D.9 Adversarial-memory control

The adversarial-memory condition deliberately constructs a memory vector anti-aligned with the true target.

The benchmark implementation computes:

$$\mathbf{m}_{adv} = -\mathbf{m}_{target} + 0.75\,\mathbf{m}_{competitor}.$$

This condition tests whether coherent reconstruction can survive under actively misleading latent structure.

Successful collapse under adversarial memory strongly supports the claim that coherent latent alignment is functionally necessary.

D.10 Bounded-memory constitutional control

The benchmark includes bounded latent arbitration using a soft clipping operator:

$$\mathbf{m}_{bounded} = \tanh\left(\frac{\mathbf{m}}{c}\right) c.$$

This mechanism constrains latent amplification while preserving dimensional continuity.

The bounded-memory regime is intended to prevent unstable explosive latent amplification from trivially dominating reconstruction.

D.11 Overdriven-arbitration instability test

The overdriven-arbitration condition intentionally exaggerates latent amplification:

$$\mathbf{m}_{over} = \tanh(4s\,\mathbf{m}).$$

This condition tests whether unconstrained arbitration destabilizes reconstruction.

The observed instability of overdriven arbitration supports the use of bounded constitutional arbitration inside the main benchmark regime.

D.12 Double-blind evaluation structure

The benchmark is designed to satisfy a computational double-blind structure:

- observable identity information is removed within alias groups;
- shuffled-label evaluation prevents direct target encoding;
- coherent and corrupted memory conditions are mixed uniformly;
- and reconstruction is evaluated only after score generation.

No benchmark mode has direct access to:

- explicit target indices;
- observable identity uniqueness;
- or external reconstruction labels.

D.13 Interpretation

The anti-cheat controls collectively support the following benchmark interpretation:

> Successful reconstruction cannot be explained solely by observable similarity, label leakage, trivial arbitration shortcuts, or random latent activation.

Instead, successful recovery requires:

- coherent latent-memory structure;
- bounded latent arbitration;
- and target-consistent latent alignment.

The anti-cheat layer is therefore treated as part of the core scientific contribution of the OMM–SBT–LIM framework itself.

E Appendix E: Code Availability

This appendix documents repository access, computational requirements, execution procedures, and reproducibility conventions associated with the OMM–SBT–LIM framework.

The objective of the appendix is to ensure that all principal benchmark results remain independently reproducible and auditable.

E.1 Public repository

The reference implementation of the OMM–SBT–LIM benchmark framework is hosted publicly at:

https://github.com/NicolasGilbertAlbertRoux/omm-sbt-lim.git

The repository is intended to contain:

- benchmark implementations;
- anti-cheat control systems;

- reconstruction pipelines;
- latent calibration scans;
- plotting scripts;
- statistical aggregation procedures;
- reproducibility utilities;
- and benchmark result exports.

The repository is versioned to permit independent rerunning of all major experiments described in the paper.

E.2 Repository organization

The repository structure is organized around reproducible experimental pipelines.

The principal directories include:

- `experiments/`
- `results/`
- `scripts/`
- `tests/`
- and configuration files for deterministic execution.

The `experiments/` directory contains executable benchmark programs.

The `results/` directory contains generated benchmark outputs, CSV summaries, and aggregated statistical tables.

The `tests/` directory contains lightweight verification tests used to validate metric consistency and reconstruction logic.

E.3 Primary benchmark programs

The repository includes executable implementations of:

- observable-alias memory-necessity experiments;
- strict double-blind anti-cheat benchmarks;
- latent calibration scans;
- shuffled-label controls;
- bounded-memory arbitration tests;
- and adversarial-memory perturbation procedures.

The strict double-blind benchmark constitutes the principal anti-cheat evaluation framework of the paper.

E.4 Execution procedure

The principal benchmark pipeline can be reproduced using:

```
python -m pip install -r requirements.txt
python -m pytest
python experiments/observable_alias_memory_necessity.py
python experiments/strict_double_blind_memory_necessity.py
python scripts/inspect_available_results.py
```

The benchmark suite was designed to execute using standard scientific Python tooling without specialized hardware acceleration requirements.

E.5 Software environment

The benchmark framework uses:

- Python 3.10;
- NumPy;
- Pandas;
- SciPy;
- Matplotlib;
- and PyTest.

Representative execution environments include:

- macOS-based systems;
- standard CPU execution;
- and deterministic pseudo-random initialization using fixed benchmark seeds.

The framework intentionally avoids dependence on:

- GPU-only kernels;
- distributed infrastructure;
- proprietary numerical backends;
- or closed external APIs.

E.6 Deterministic reproducibility

Benchmark reproducibility is enforced through:

- fixed pseudo-random seeds;
- deterministic alias-group construction;
- deterministic latent-vector generation;
- deterministic benchmark grids;
- and explicit parameter definitions.

Whenever stochastic tie-breaking occurs, it is controlled through seeded pseudo-random generators.

E.7 Exported outputs

The repository exports benchmark outputs including:

- raw reconstruction tables;
- CSV summary statistics;
- dominance measurements;
- recovery statistics;
- false-positive rates;
- anti-cheat verdict summaries;
- and aggregated interpretation files.

Representative exported files include:

- `raw_results.csv`
- `summary_by_mode.csv`
- `summary_by_condition_mode.csv`
- and `interpretation.json`

E.8 Verification tests

The repository contains lightweight automated tests intended to verify:

- metric consistency;
- reconstruction scoring behavior;
- normalization correctness;
- and benchmark pipeline integrity.

The test suite is intentionally minimal to preserve portability and rapid verification.

E.9 Computational scale

The strict double-blind benchmark explores a large combinatorial search space involving:

- alias groups;
- reconstruction candidates;
- memory conditions;
- arbitration speeds;
- constitutional bounds;
- and anti-cheat perturbation regimes.

Representative executions may evaluate tens of millions of scored reconstruction jobs.

Progress indicators and execution statistics are therefore included to support long-duration auditability.

E.10 Scientific auditability

The repository is intended not merely as supplementary code, but as an experimental audit layer for the paper itself.

The principal scientific objective is independent falsifiability.

Accordingly, the framework is designed to permit:

- independent rerunning of experiments;
- independent anti-cheat inspection;
- independent parameter modification;
- independent stress testing;
- and independent adversarial evaluation.

E.11 Scope limitations

The repository does not claim:

- biological realism;
- neural completeness;
- metaphysical proof;
- or direct consciousness simulation.

The codebase implements a constrained formal reconstruction framework intended to evaluate latent-memory necessity under strict observable aliasing conditions.

E.12 Long-term archival objective

Because reproducibility is treated as a central epistemic requirement of the framework, long-term archival preservation of:

- benchmark scripts;
- result tables;
- parameter grids;
- and anti-cheat procedures

is considered scientifically important.

The repository therefore functions both as:

- a computational supplement;
- and an auditable scientific record of the benchmark framework.

F Appendix F: Anatomical Notes

This appendix summarizes the principal anatomical and neurofunctional considerations that motivated portions of the conceptual architecture of the OMM–SBT–LIM framework.

The appendix is intentionally interpretative rather than demonstrative.

No direct biological equivalence between the benchmark framework and the human nervous system is claimed.

Instead, the anatomical discussion serves only as a heuristic bridge between:

- observable sensory integration,
- latent arbitration,
- persistent internal state,
- and constrained reconstruction behavior.

F.1 General caution

The present framework does not claim:

- a full neural theory of consciousness;
- a complete model of brain function;
- or biological simulation fidelity.

The benchmark architecture is a constrained mathematical formalism.

Any anatomical analogy must therefore be interpreted cautiously.

The role of the appendix is not to establish neuroscientific proof, but to identify possible functional correspondences worthy of future investigation.

F.2 Reticular formation and global regulation

The reticular formation is a distributed network extending throughout the brainstem and is associated with:

- arousal regulation;
- sleep-wake transitions;
- autonomic modulation;
- attentional gating;
- and large-scale state coordination.

Historically, lesions affecting the ascending reticular activating system have been associated with profound disturbances of wakefulness and consciousness.

Within the present framework, the reticular system is interpreted as a possible biological analogue of global arbitration constraints.

In particular, the reticular system may be viewed as participating in:

- state stabilization;
- selective amplification;
- and bounded activation propagation.

This interpretation loosely parallels the benchmark notion of bounded latent arbitration.

F.3 Thalamic integration

The thalamus occupies a central position in large-scale cortical integration.

It participates in:

- sensory relay;
- attentional modulation;
- oscillatory coordination;
- and distributed cortico-thalamic synchronization.

Several theories of consciousness assign major integrative importance to thalamo-cortical dynamics.

Within OMM–SBT–LIM, the thalamus can be interpreted heuristically as a candidate structure for:

- latent routing;
- arbitration coordination;
- and distributed representational binding.

The benchmark does not simulate thalamic circuitry directly.

However, the conceptual role of latent arbitration bears partial functional resemblance to integrative coordination mechanisms often attributed to thalamo-cortical systems.

F.4 Raphe nuclei and modulatory persistence

The raphe nuclei constitute a major serotonergic system projecting widely throughout the brain.

These nuclei participate in:

- global neuromodulation;
- affective regulation;
- sleep dynamics;
- behavioral persistence;
- and state-dependent processing.

From the perspective of the present framework, widespread modulatory systems are of particular conceptual interest because they influence:

- persistence,
- stability,
- and global reconstruction context

rather than merely localized sensory content.

The latent-memory layer of the benchmark may therefore be interpreted as functionally closer to distributed modulatory state than to isolated sensory encoding.

F.5 Distributed reconstruction versus local encoding

One of the conceptual motivations of the framework is the distinction between:

- local observable encoding;
- and distributed latent reconstruction.

Many biological systems appear to rely heavily on distributed recurrent integration rather than simple feedforward sensory registration.

Examples include:

- predictive coding dynamics;
- recurrent cortical processing;
- distributed working memory;
- and global state-dependent modulation.

The strict observable-alias benchmark was partly designed to formalize a similar distinction:

$$\text{observable similarity} \neq \text{identity reconstruction.}$$

F.6 Persistent identity and state continuity

A central conceptual question underlying the benchmark concerns the relationship between:

- transient observables;
- and persistent identity continuity.

Biological nervous systems exhibit:

- long-timescale memory persistence;
- recurrent self-referential dynamics;
- and state continuity across changing sensory conditions.

The benchmark does not claim to reproduce these phenomena directly.

However, the necessity of coherent latent structure under strict observable aliasing may be interpreted as mathematically compatible with the broader idea that persistence cannot always be reduced to immediate observable content alone.

F.7 State-space interpretation

The benchmark framework can also be interpreted geometrically.

Observable vectors occupy one representational manifold:

$$\mathcal{O},$$

while latent-memory vectors occupy another:

$$\mathcal{M}.$$

Under strict aliasing:

$$\exists i \neq j \quad \text{such that} \quad \mathbf{o}_i = \mathbf{o}_j,$$

while:

$$\mathbf{m}_i \neq \mathbf{m}_j.$$

The reconstruction problem therefore becomes impossible inside observable space alone.

This separation loosely resembles the distinction between:

- immediate sensory equivalence;
- and internally differentiated system state.

F.8 Consciousness-related interpretations

Some speculative interpretations of consciousness emphasize:

- global integration;
- recurrent arbitration;
- persistent latent structure;
- and distributed coordination.

The present framework does not validate any specific theory of consciousness.
In particular, it does not establish:

- subjective experience;
- phenomenology;
- selfhood;
- or qualia.

The benchmark merely demonstrates that, within the formal regime constructed here, latent structure becomes mathematically necessary for identity reconstruction under strict observable ambiguity.

F.9 Limitations of the anatomical analogy

The anatomical parallels discussed in this appendix remain speculative and incomplete.
The benchmark omits:

- biological plasticity;
- spiking dynamics;
- metabolic constraints;
- developmental structure;
- embodied interaction;
- and large-scale neural heterogeneity.

The framework should therefore not be interpreted as a direct biological model.
Its principal contribution remains mathematical and epistemic rather than neuroscientific.

F.10 Interpretative role of the appendix

The purpose of the appendix is ultimately conceptual.

It situates the OMM–SBT–LIM framework within a broader landscape of questions concerning:

- latent structure;
- persistence;
- distributed arbitration;
- and observable insufficiency.

Whether future neuroscience will support, reject, refine, or reinterpret these analogies remains an open empirical question.

G Appendix G: Public Explanation Outside the Scientific Core

This appendix is intentionally separated from the scientific core of the paper. It presents an interpretative and public-facing explanation of the conceptual motivations behind the OMM–SBT–LIM framework. The appendix should not be interpreted as constituting part of the formal empirical claims established by the benchmark itself.

G.1 Why this framework was developed

The central question motivating the OMM–SBT–LIM framework can be expressed informally as follows:

> Can identity be reconstructed purely from what is immediately observable, or does reconstruction require an additional latent internal structure?

The benchmark was designed to formalize this question under strict mathematical constraints.

Rather than debating philosophical intuitions directly, the framework attempts to build a falsifiable reconstruction environment in which the problem can be tested computationally.

G.2 The intuition behind observable aliasing

In ordinary life, many systems appear identifiable because observable differences remain available.

However, the situation changes radically if observables become indistinguishable.

The benchmark therefore constructs a deliberately difficult regime: multiple identities share the exact same observable representation.

Under such conditions, observable appearance alone becomes insufficient for discrimination.

The framework then asks whether coherent latent structure can still permit reconstruction.

G.3 Why anti-cheat controls matter

A major difficulty in reconstruction research is hidden information leakage.

Systems often appear to succeed because:

- labels leak implicitly;

- observables accidentally encode identity;
- or shortcuts emerge unnoticed.

For this reason, the benchmark incorporates multiple anti-cheat layers:

- observable aliasing;
- shuffled-label controls;
- adversarial memory;
- destroyed memory;
- memory swaps;
- and null-memory baselines.

The goal is not merely performance, but falsifiability.

G.4 What the benchmark actually shows

Within the formal conditions constructed in the benchmark, coherent identity reconstruction becomes impossible from observables alone once strict aliasing is imposed.

Successful reconstruction instead requires:

- coherent latent structure;
- bounded arbitration;
- and internally consistent memory alignment.

This result does *not* establish metaphysical conclusions.

It does not prove:

- consciousness;
- souls;
- subjective experience;
- or any supernatural interpretation.

The result is purely structural: under these formal conditions, observable information alone is insufficient.

G.5 Why the result may still matter

Even without metaphysical implications, the result may still carry scientific and philosophical importance.

Many discussions concerning:

- identity,
- memory,
- continuity,
- cognition,

- and consciousness

implicitly assume that observable state fully determines reconstructible identity.

The benchmark challenges that assumption within a strict mathematical regime.

It therefore suggests that:

> observable equivalence does not necessarily imply reconstructive equivalence.

G.6 Why the framework remains cautious

The framework intentionally avoids exaggerated conclusions.

The benchmark is:

- simplified;
- abstract;
- non-biological;
- and mathematically constrained.

The paper therefore does not claim:

- to solve consciousness;
- to reproduce the human mind;
- or to establish a final theory of identity.

Instead, the work proposes a formal reconstruction regime whose behavior can be independently tested, criticized, modified, or falsified.

G.7 Scientific humility

Historically, many scientific advances began with simplified formal systems that initially appeared disconnected from reality.

At the same time, many speculative ideas failed completely.

The present framework may ultimately:

- prove useful;
- prove incomplete;
- require major revision;
- or be entirely superseded.

For this reason, the framework is presented cautiously and with explicit anti-cheat transparency.

G.8 The role of reproducibility

A central principle of the project is that extraordinary conceptual claims require reproducible computational transparency.

The repository therefore exists not merely for openness, but for independent auditability. Anyone should be able to:

- rerun the benchmarks;
- modify the controls;
- test adversarial variants;
- and attempt falsification independently.

This emphasis on reproducibility is treated as part of the scientific content itself.

G.9 Possible future directions

Future work could explore:

- larger latent spaces;
- adaptive arbitration dynamics;
- recurrent architectures;
- biological constraints;
- probabilistic memory degradation;
- temporal reconstruction;
- and distributed multi-agent regimes.

Whether such extensions would strengthen or weaken the present findings remains an open empirical question.

G.10 Closing perspective

The OMM–SBT–LIM framework should ultimately be interpreted neither as a metaphysical doctrine nor as a complete theory of mind.

It is better understood as:

> a constrained scientific attempt to test whether latent structure can become formally necessary for reconstruction once observable information is intentionally made insufficient.

If the framework proves useful, its value will come not from rhetorical claims, but from:

- falsifiability,
- reproducibility,
- anti-cheat rigor,
- and independent verification.

Those principles, rather than any single interpretation, constitute the core intellectual commitment of the project.